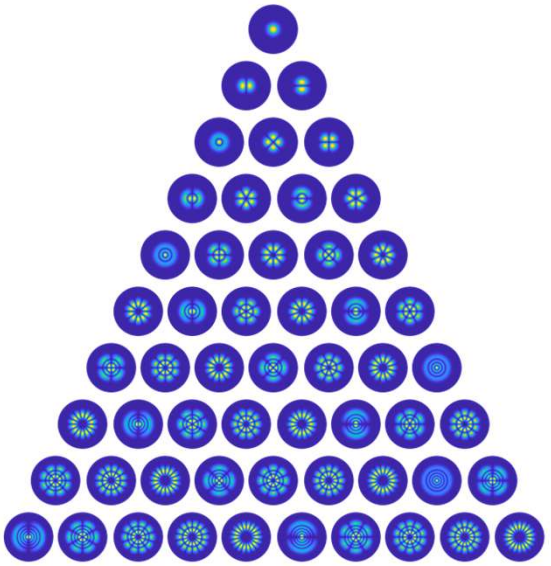
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

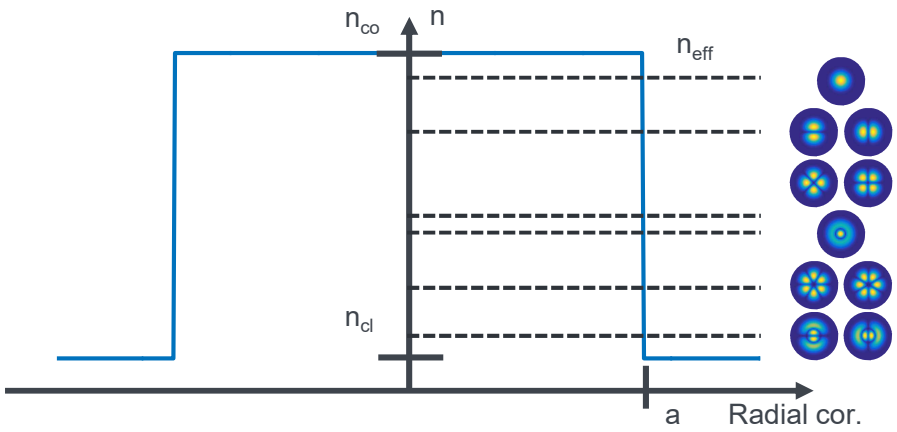
Lecture 7

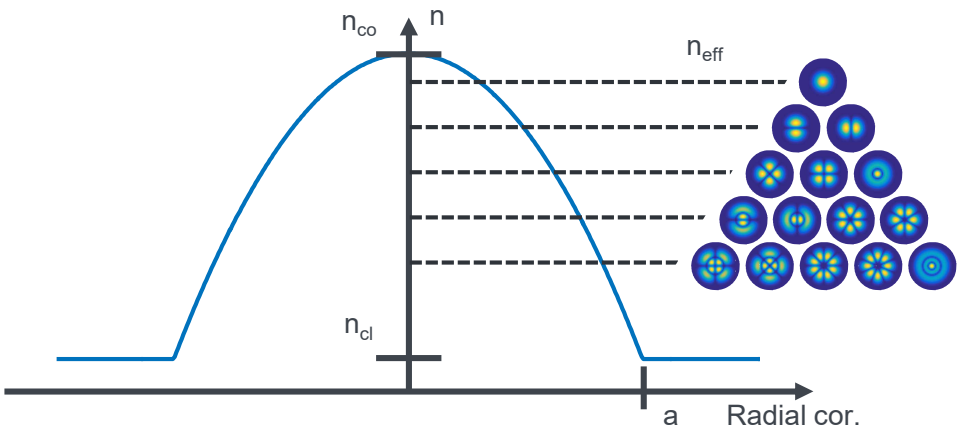


Prof. Dr.-Ing. Georg Rademacher

Multi-mode fibers (MMFs)

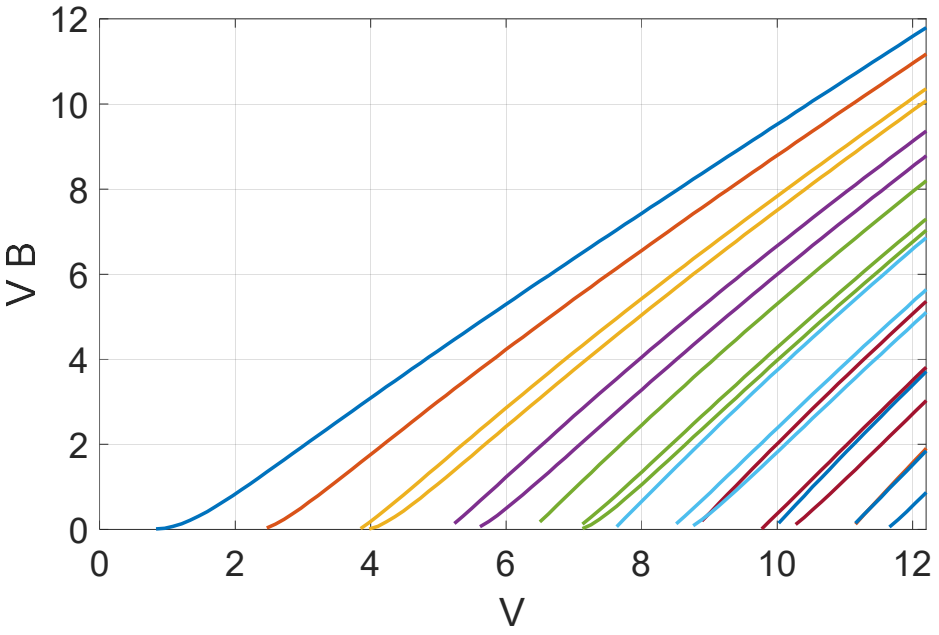
University of Stuttgart
Germany

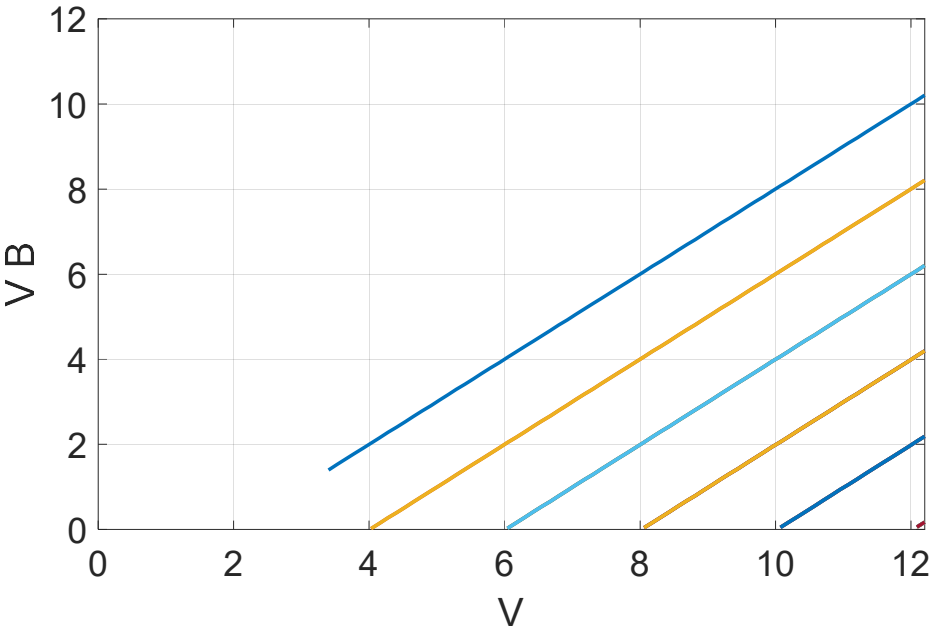




Graded-index fibers

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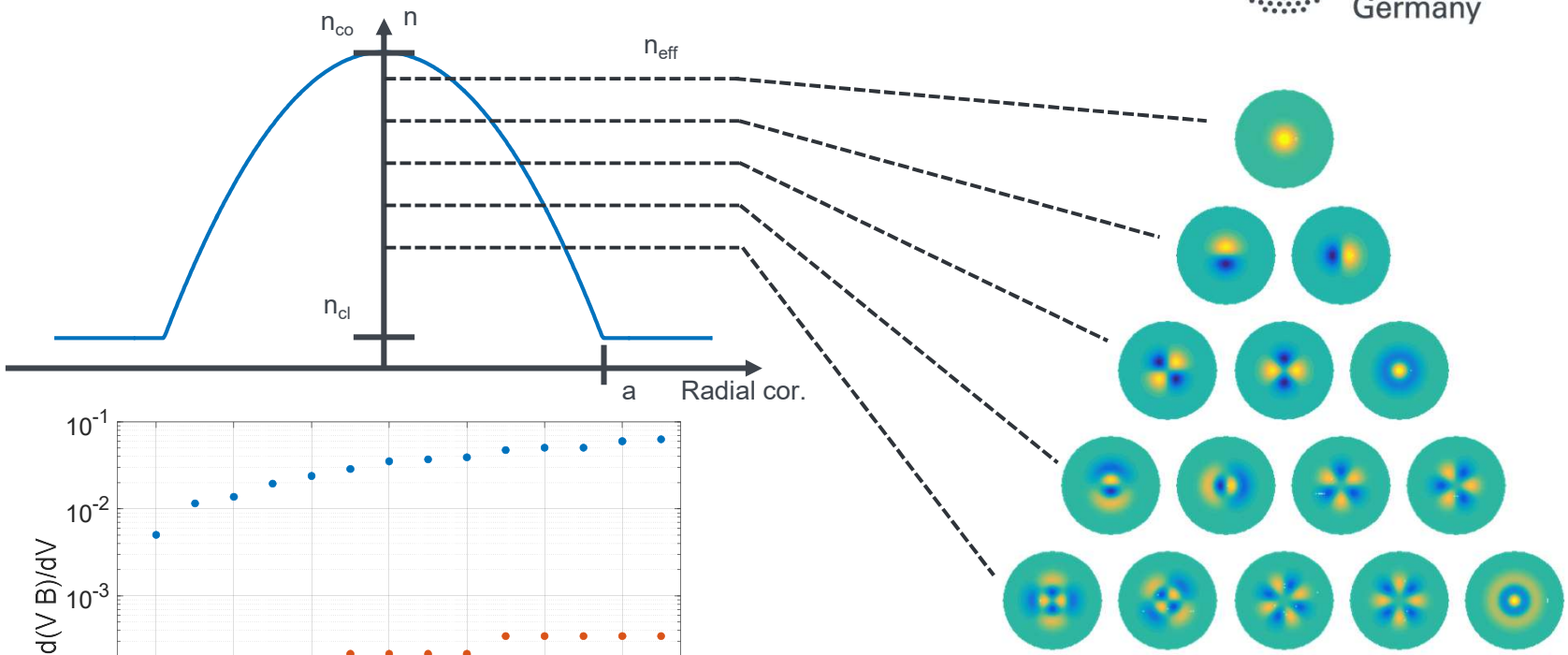


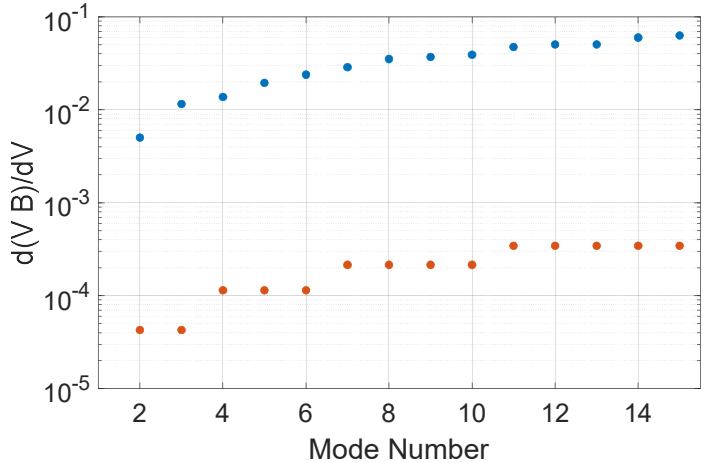


Multi-mode fibers (MMFs)

- Nomenklatur!

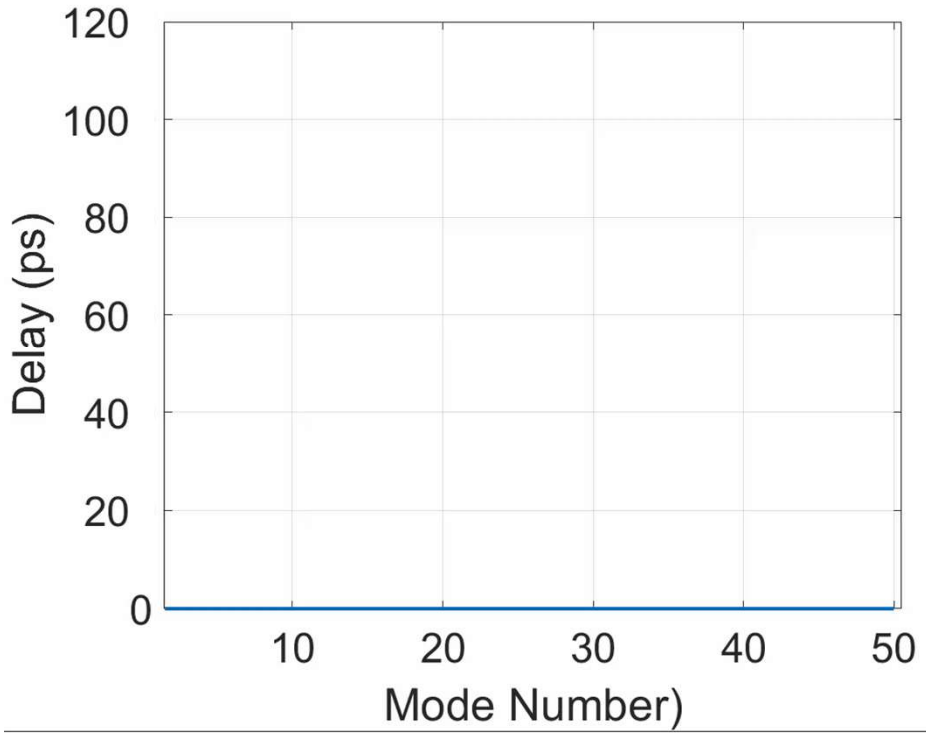
University of Stuttgart
Germany

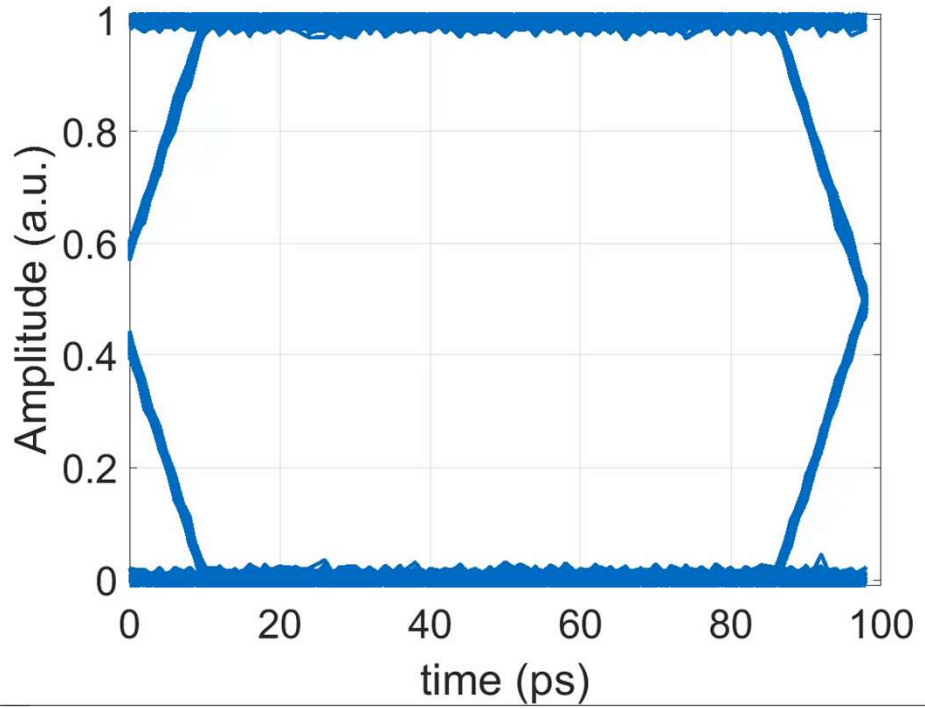




Why is the delay difference so important?

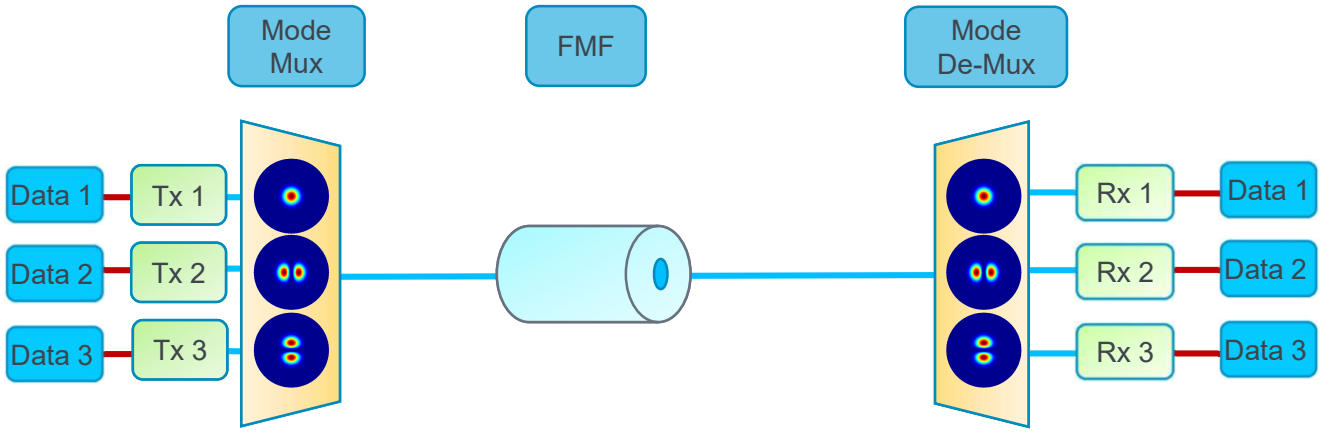
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Germany





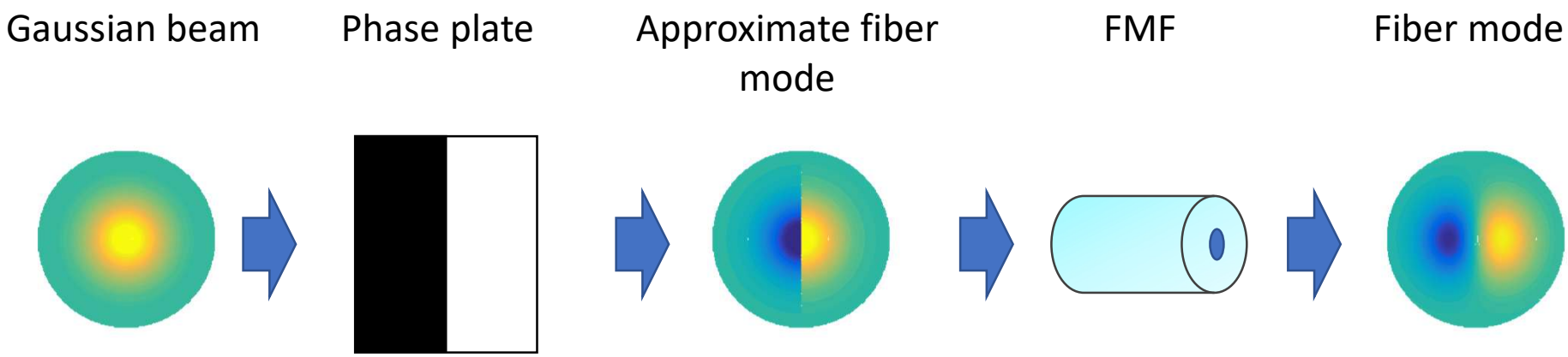
Space division multiplex with MMF

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Germany



Mode-Multiplexers

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- Draw how to make 3-mode device with phase plates, mirrors and beam splitter

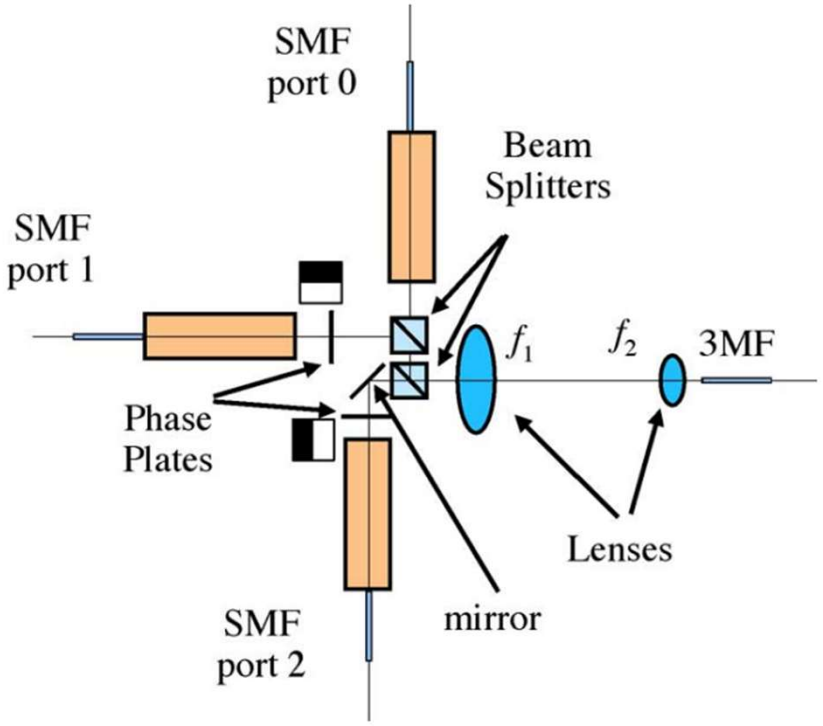
JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 30, NO. 4, FEBRUARY 15, 2012

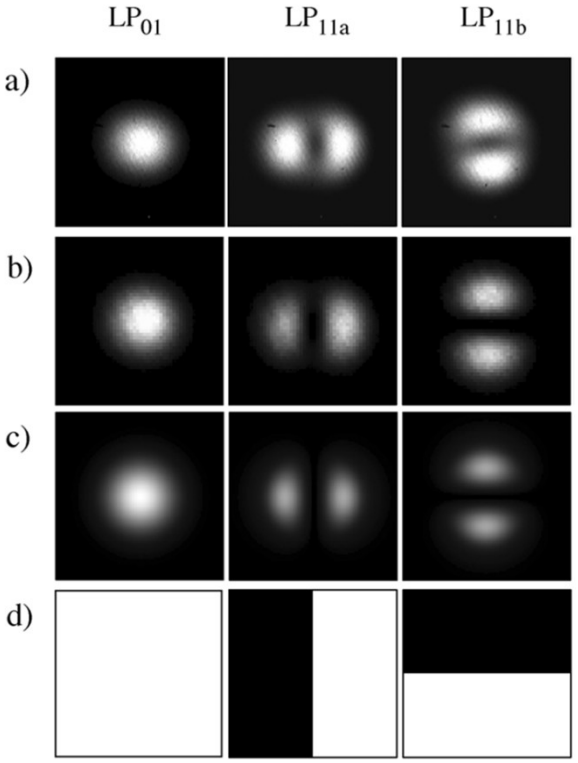
Mode-Division Multiplexing Over 96 km of Few-Mode Fiber Using Coherent 6×6 MIMO Processing

Roland Ryf, *Member, IEEE*, Sebastian Randel, *Senior Member, IEEE*, Alan H. Gnauck, *Fellow, IEEE, Fellow, OSA*, Cristian Bolle, Alberto Sierra, *Student Member, IEEE*, Sami Mumtaz, Mina Esmaeelpour, Ellsworth C. Burrows, René-Jean Essiambre, *Senior Member, IEEE, Fellow, OSA*, Peter J. Winzer, *Fellow, IEEE, Member, OSA*, David W. Peckham, *Senior Member, IEEE*, Alan H. McCurdy, *Member, IEEE*, and Robert Lingle, Jr., *Member, IEEE*

(Invited Tutorial)

Mode-Multiplexers





nature COMMUNICATIONS

ARTICLE

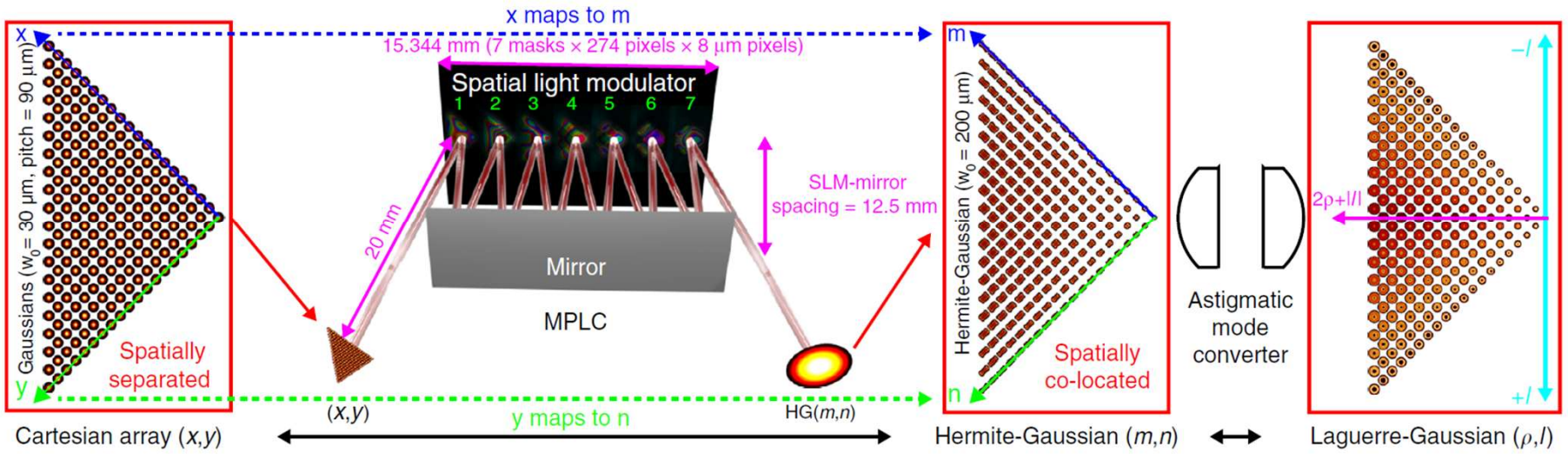
https://doi.org/10.1038/s41467-019-09840-4 OPEN

Laguerre-Gaussian mode sorter

Nicolas K. Fontaine[1,3], Roland Ryf[1], Haoshuo Chen[1], David T. Neilson[1], Kwangwoong Kim[1] & Joel Carpenter[2,3]

Multi-plane light conversion

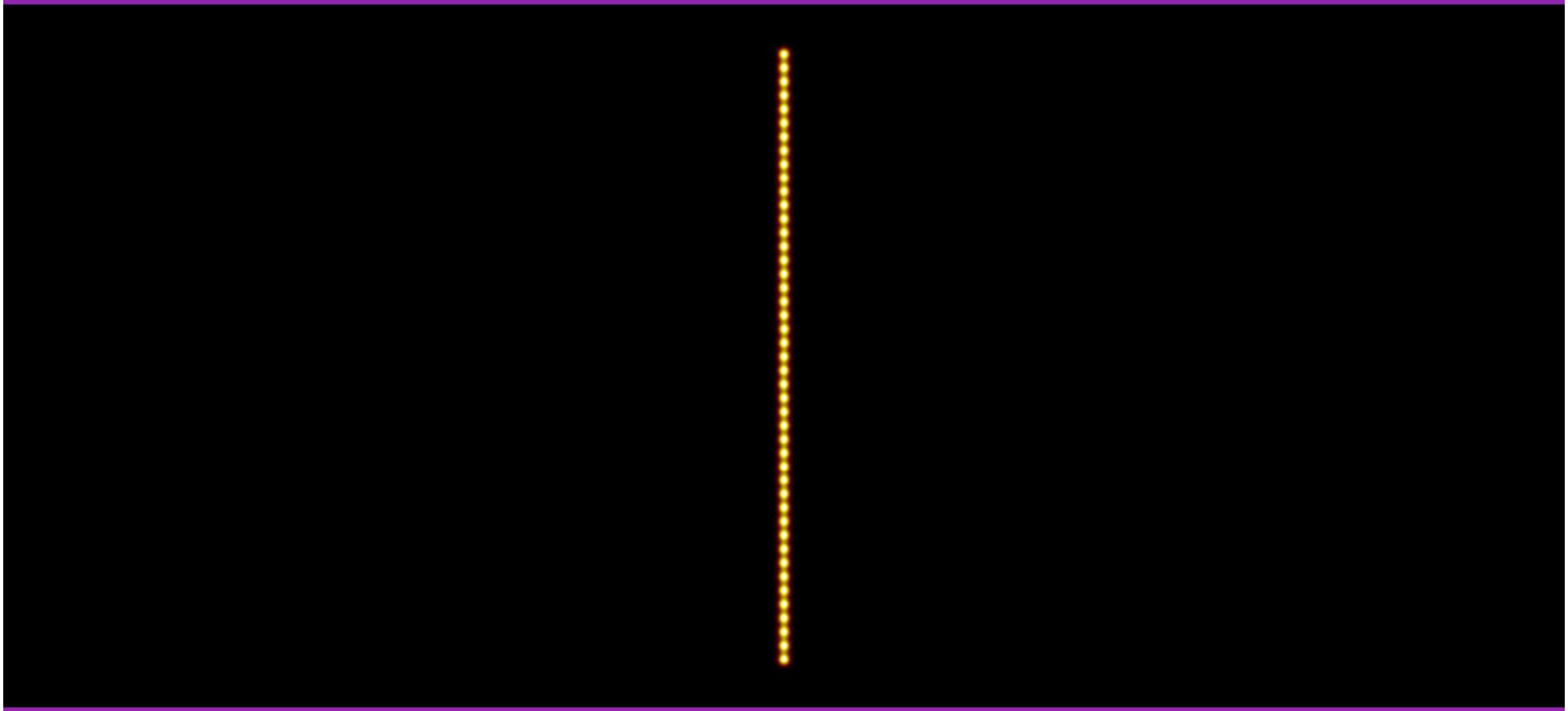
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Germany



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AUSTRALIA

Multi-plane light conversion

NOKIA Bell Labs

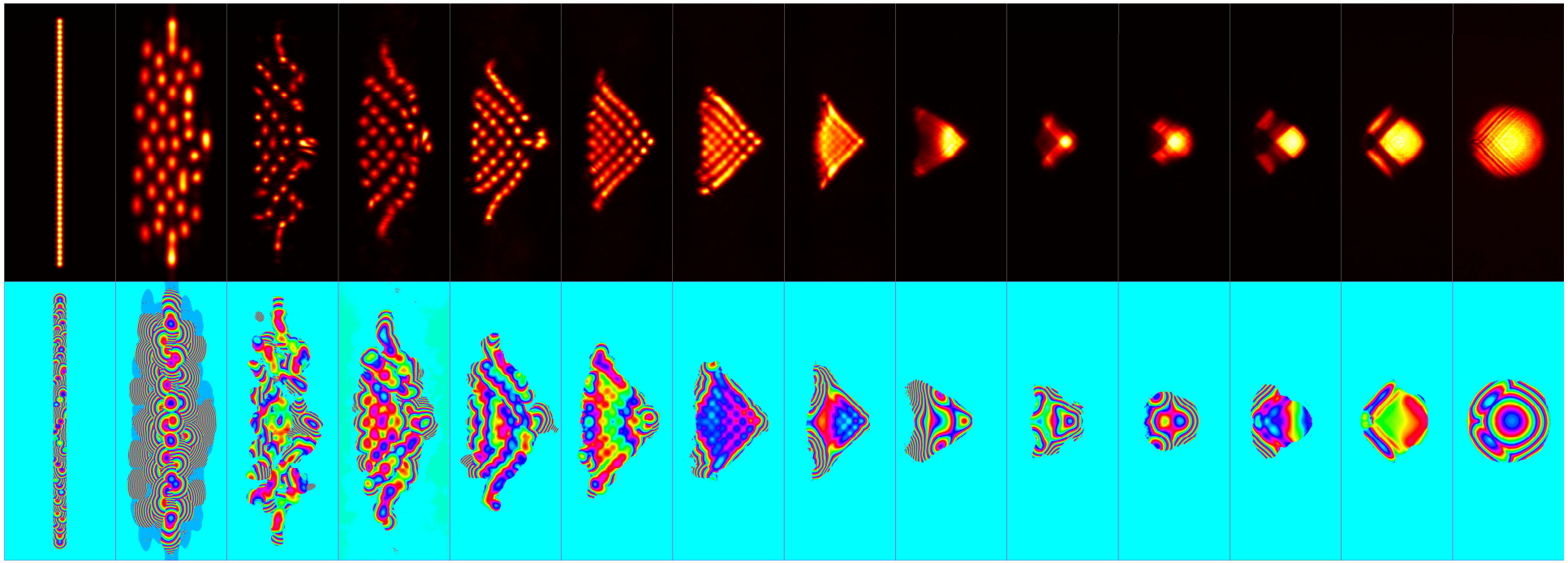


J.-F Morizur et al., J. Opt. Soc. Am. A, 27 (11) 2524-2531 (2010).

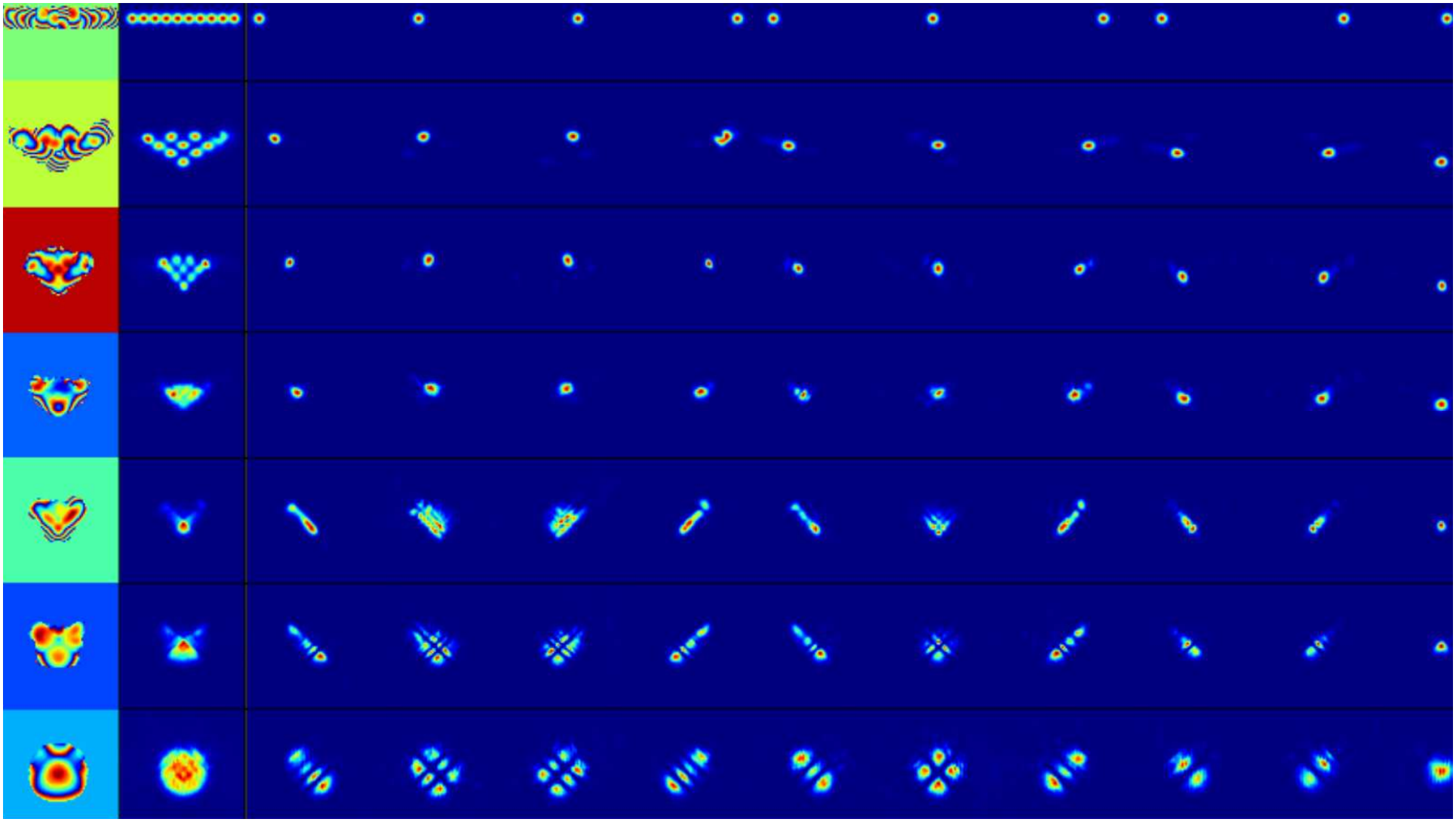
THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Triangular → HG

NOKIA Bell Labs

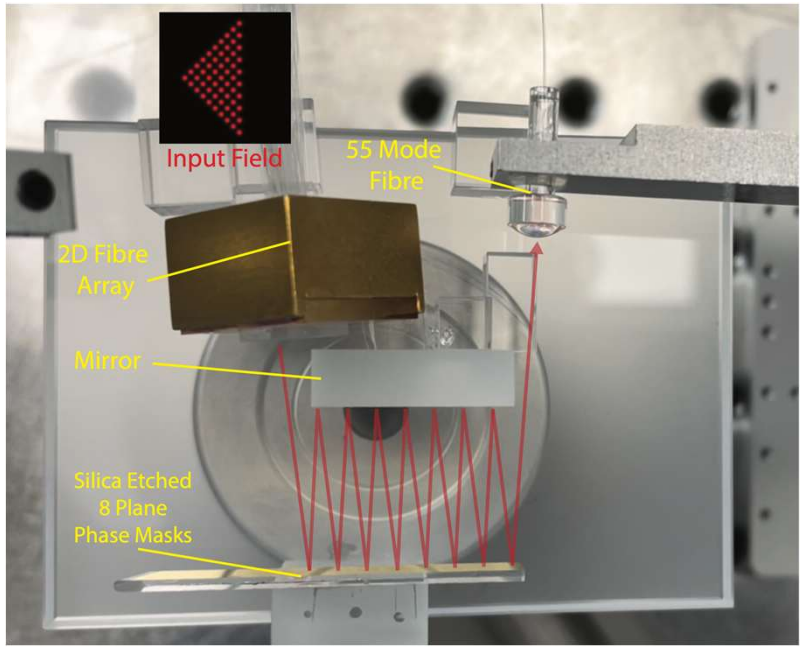


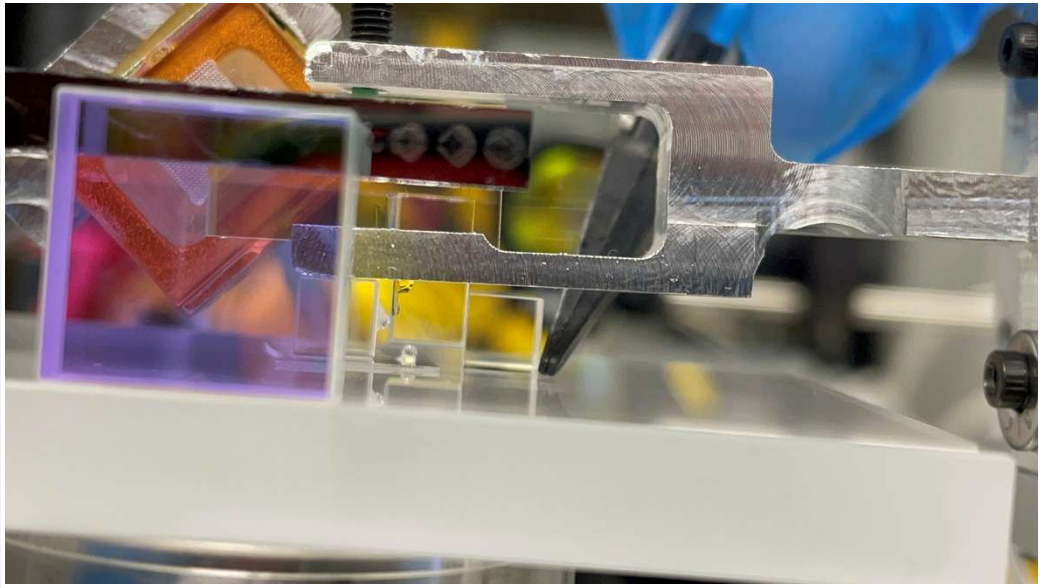
45 modes, 14 planes (512×1280 5μm pixels @ 1550nm, 24mm plane spacing)



Mode-Multiplexers

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Germany

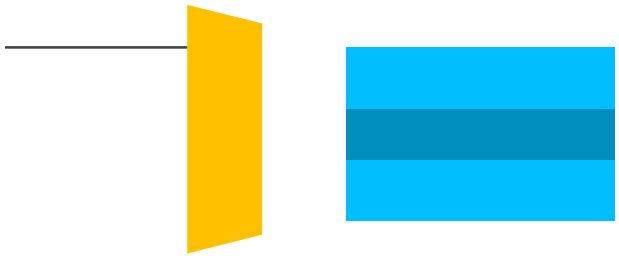


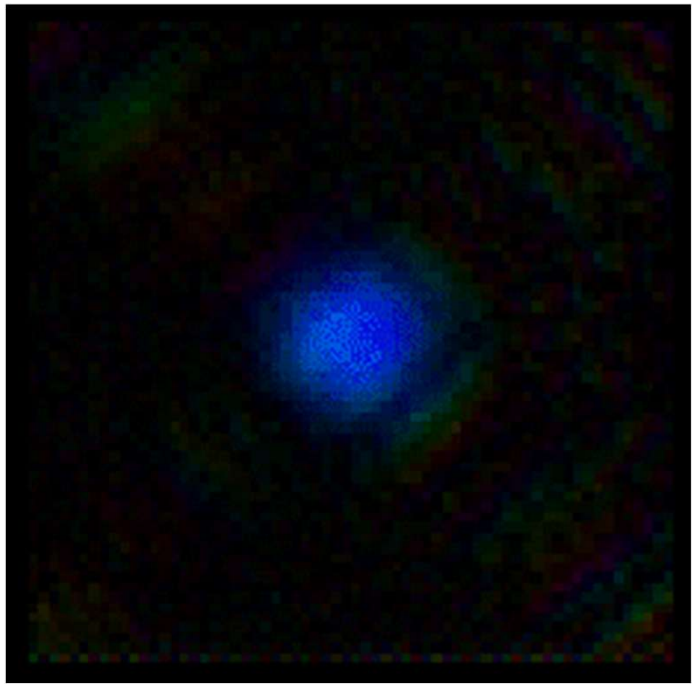


Mode-Multiplexers

University of Stuttgart
Germany

- SMF
- MUX
- MMF





Mode-Multiplexers

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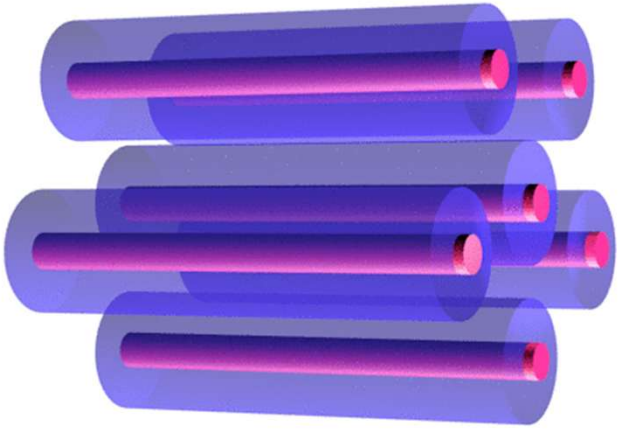


Photonic Lanterns, 3-D Waveguides, Multiplane Light Conversion, and Other Components That Enable Space-Division Multiplexing

This article reviews the mode-shaping technologies needed to perform mode-division multiplexing on few-mode fibers and discusses implementations and tradeoffs of four different techniques: multiplane light conversion, fused fiber devices, 3-D waveguides in glass, and free-space imaging systems.

By Nicolas K. Fontaine, *Senior Member IEEE*, Joel Carpenter, *Member IEEE*, Simon Gross, Sergio Leon-Saval, Yongmin Jung, David J. Richardson, *Fellow IEEE*, and Rodrigo Amezcua-Correa

Mode-Multiplexers: Photonic Lantern





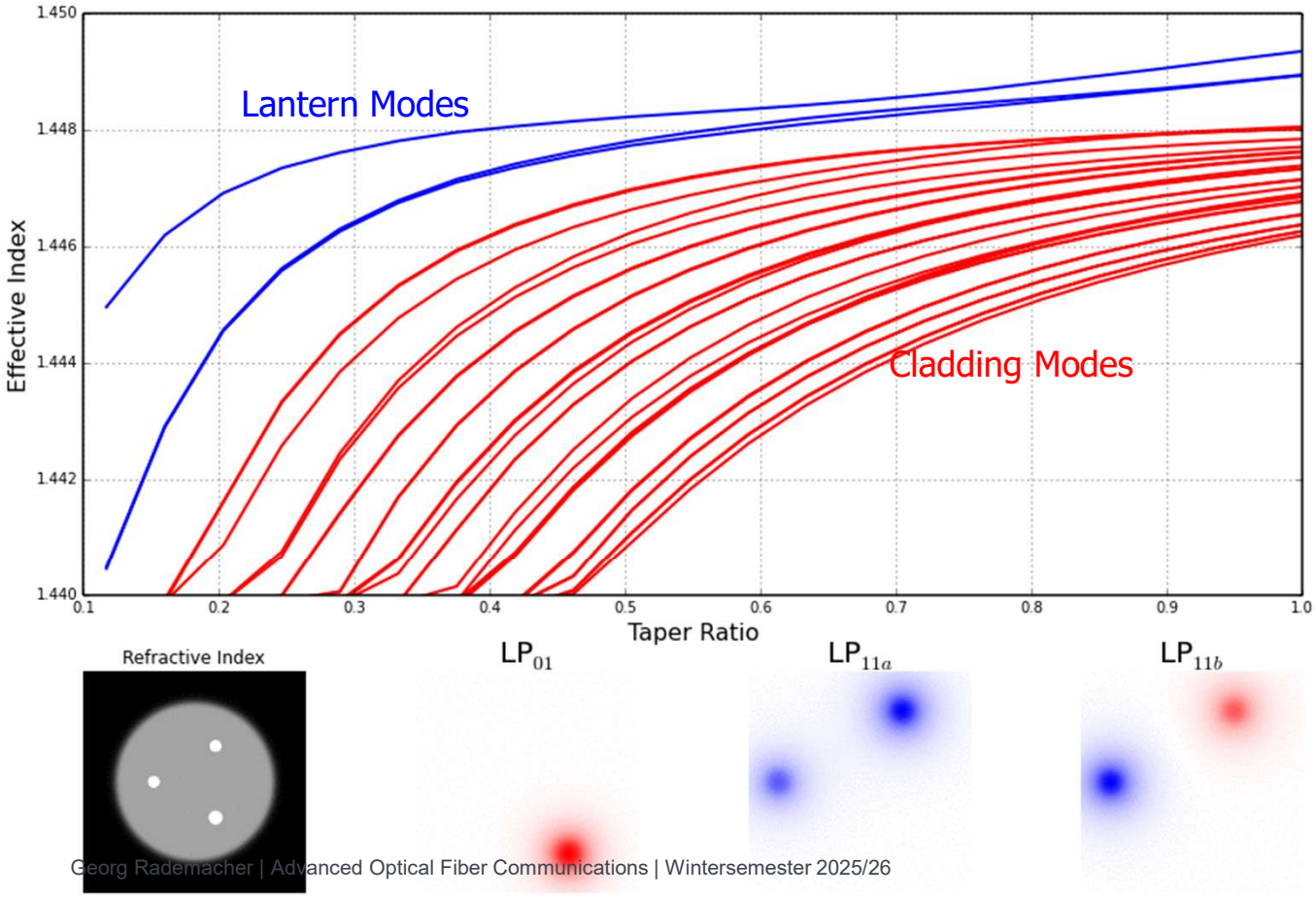
issimilar Fiber Mode Selective Lantern

diabatic transition from groups of cores to modes.

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rmany

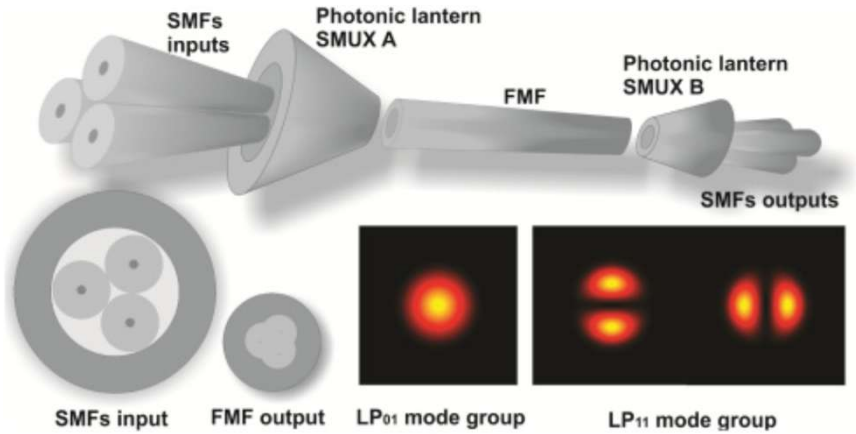
01 Big Core

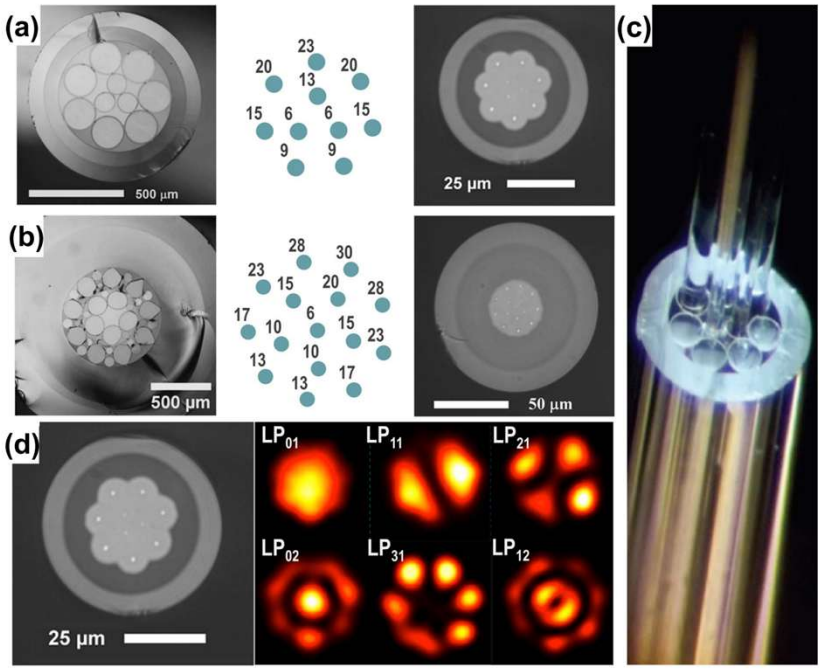
11 Two Smaller Cores

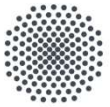


Mode-Multiplexers

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Nanophotonics 2015; 4:332–352

DE GRUYTER OPEN

Review Article

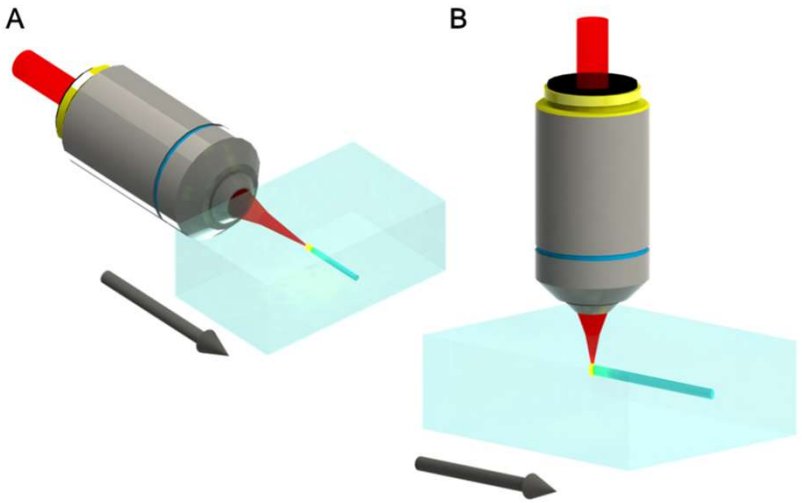
Open Access

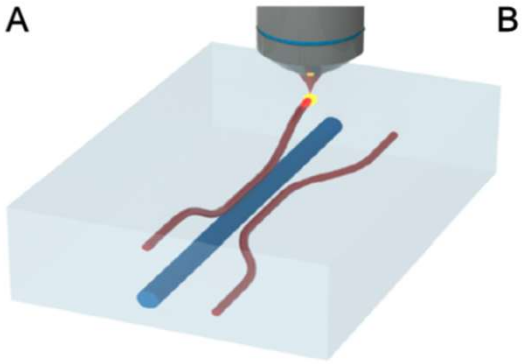
S. Gross* and M. J. Withford

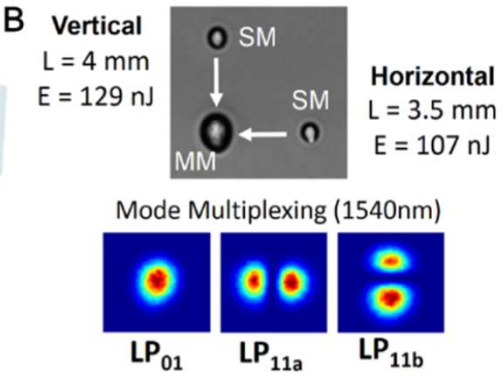
Ultrafast-laser-inscribed 3D integrated photonics: challenges and emerging applications

Mode-Multiplexers

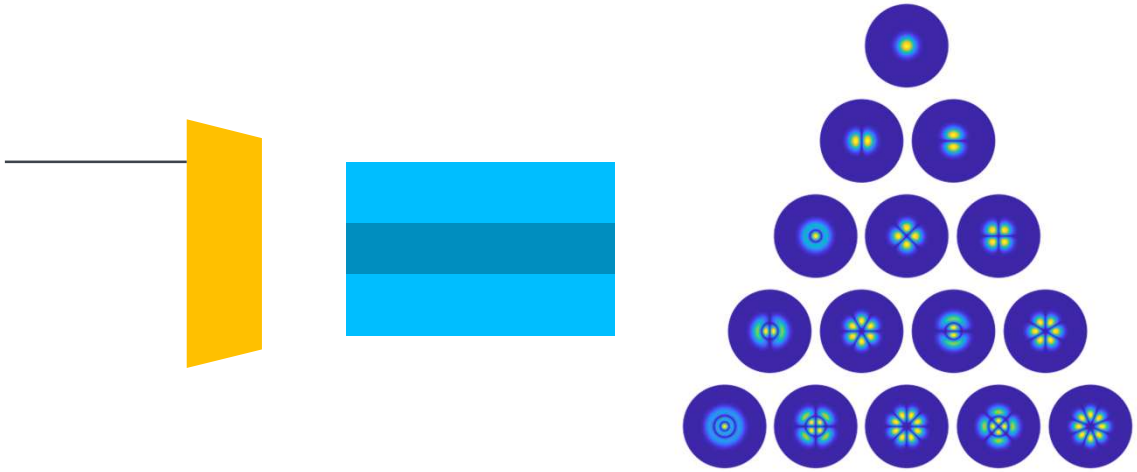
University of Stuttgart
Germany

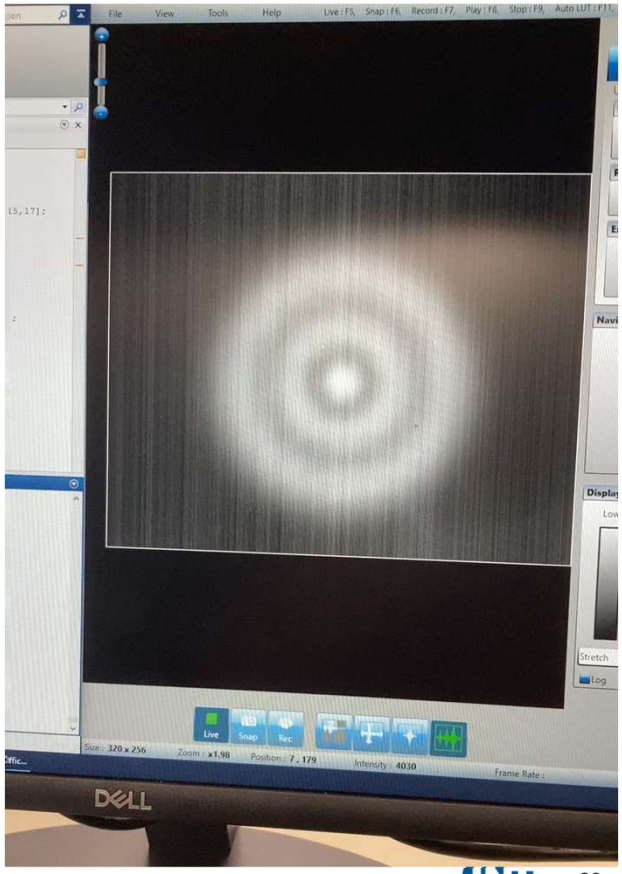




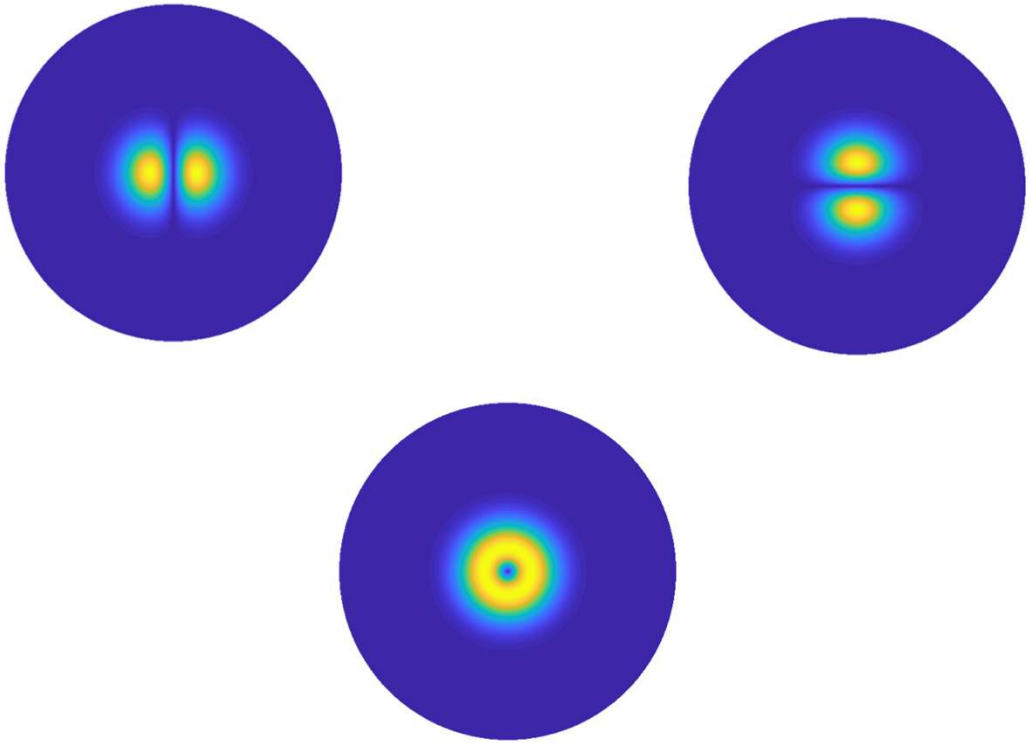


Graded-index fibers





Mode Coupling



Connecting optical fibers:

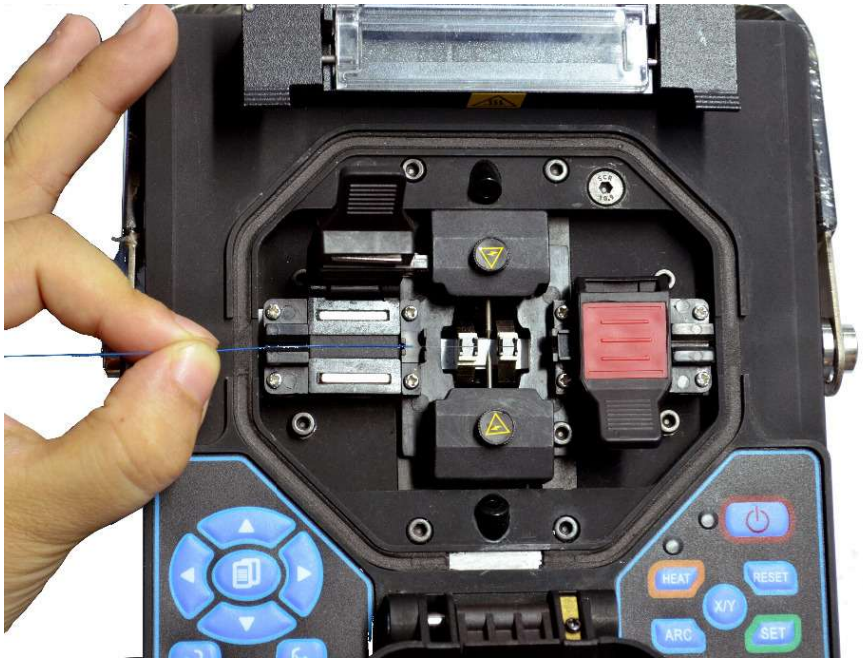
University of Stuttgart
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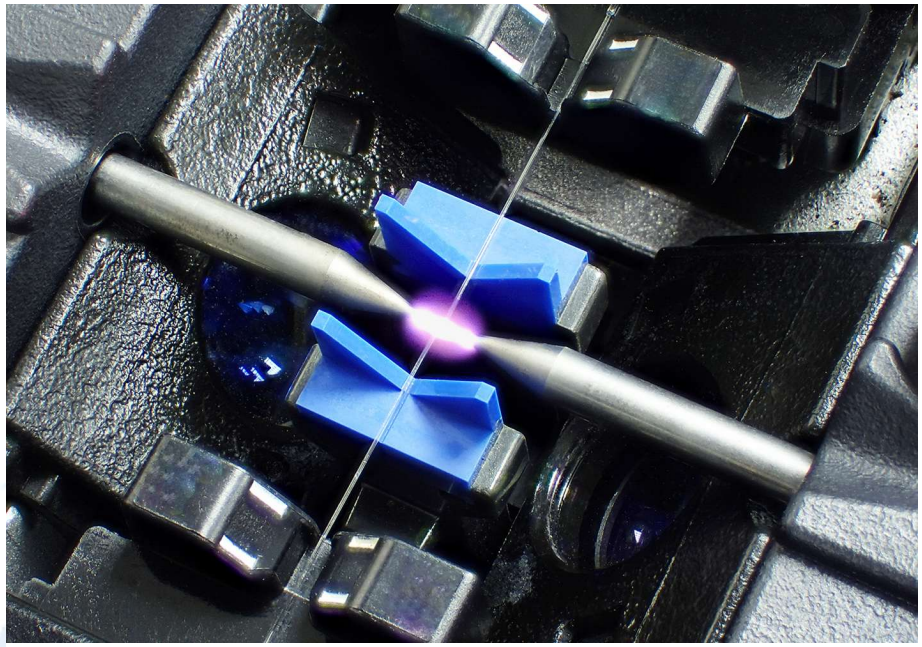




Connecting optical fibers: fusion splicing

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Germany





- Loss below 0.1 dB

Butt-coupling of SI MMFs



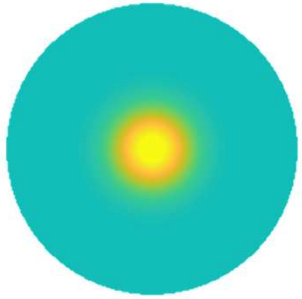
Butt-coupling of SI MMFs

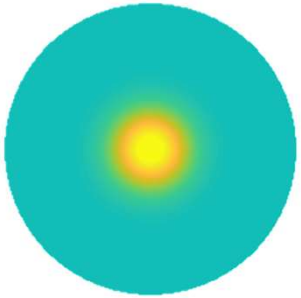


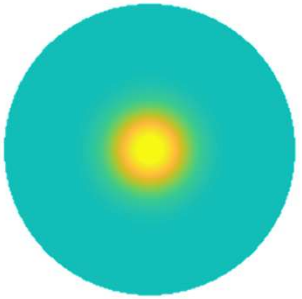
University of Stuttgart
Germany

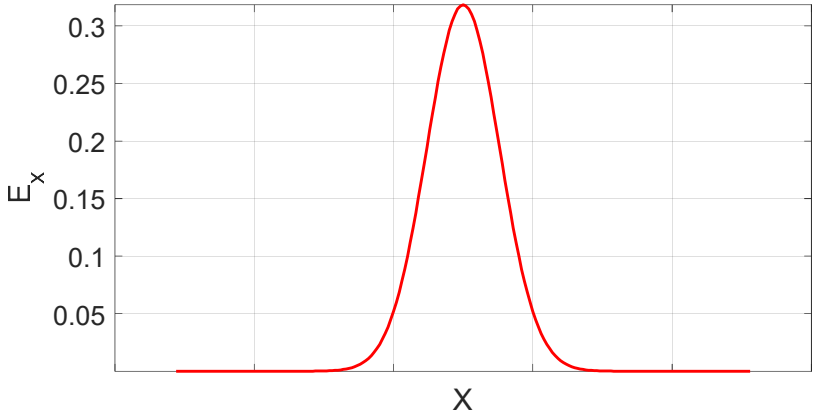
Butt-coupling of SI MMFs

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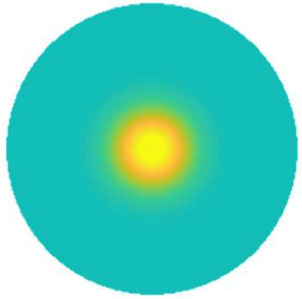


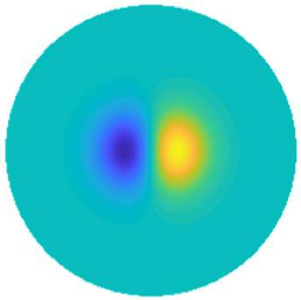




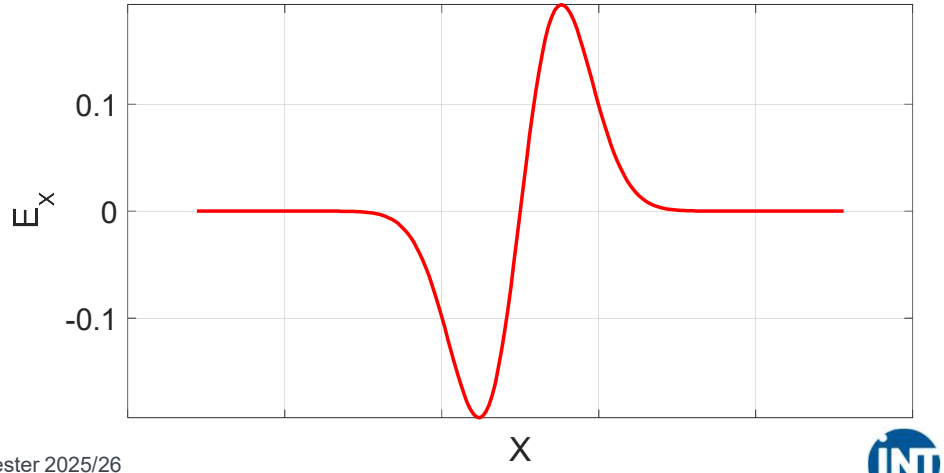
Butt-coupling of SI MMFs

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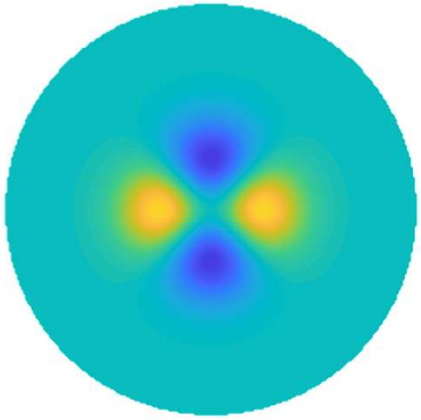


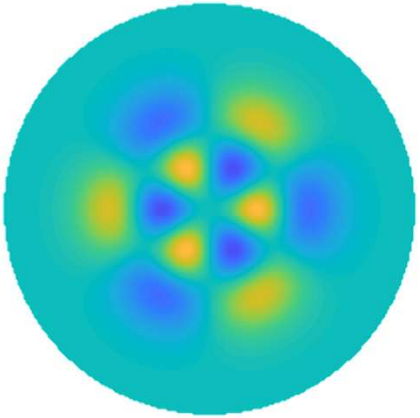


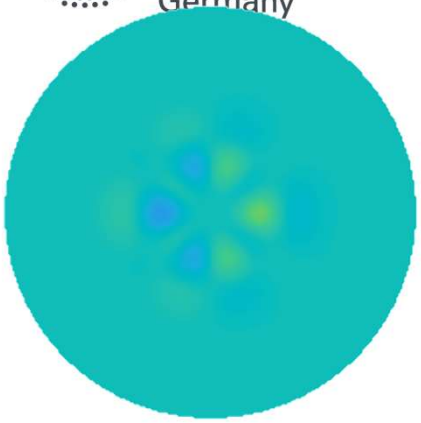


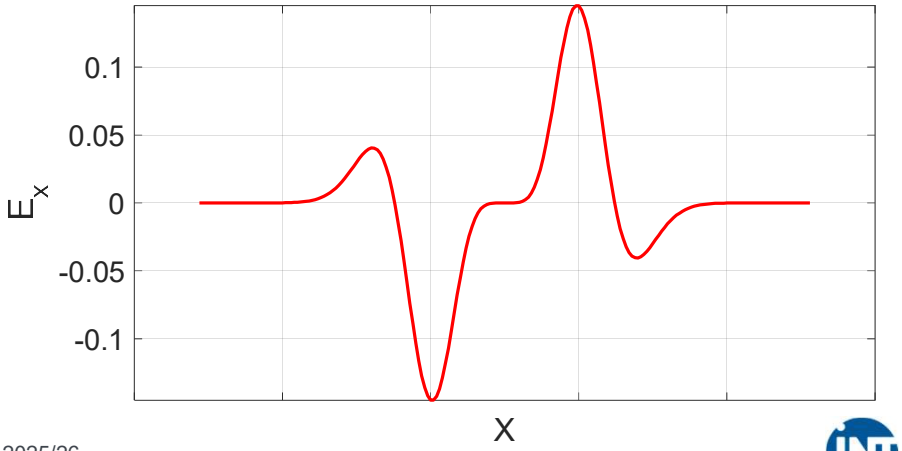


Butt-coupling of SI MMFs



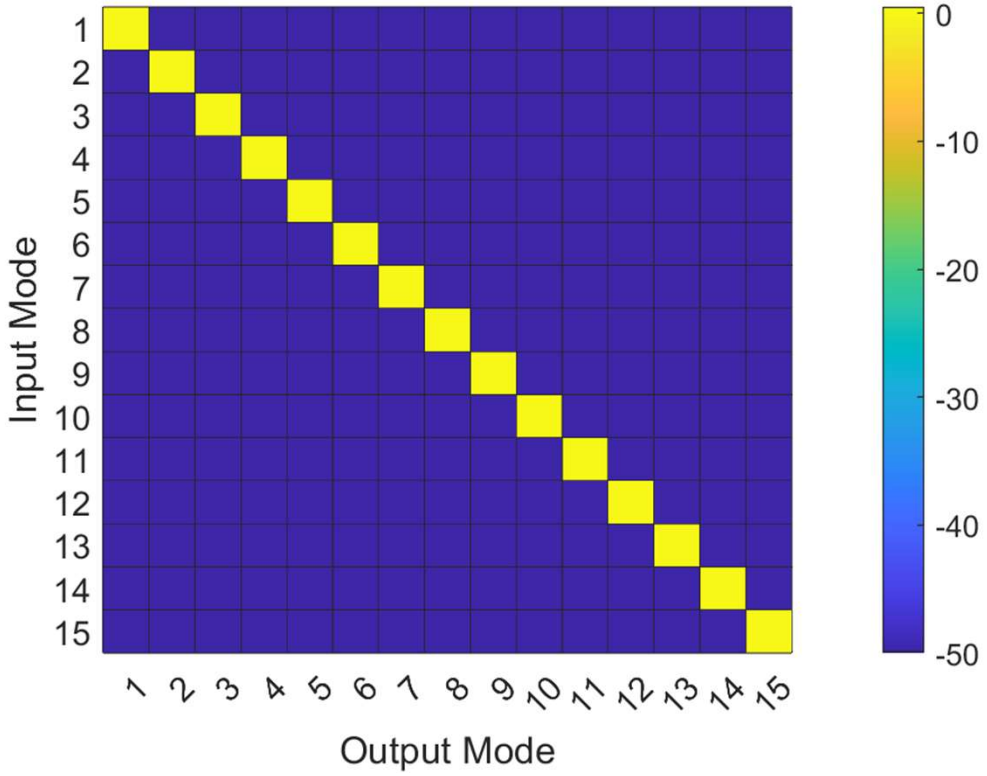






Simulation: Coupling coefficients

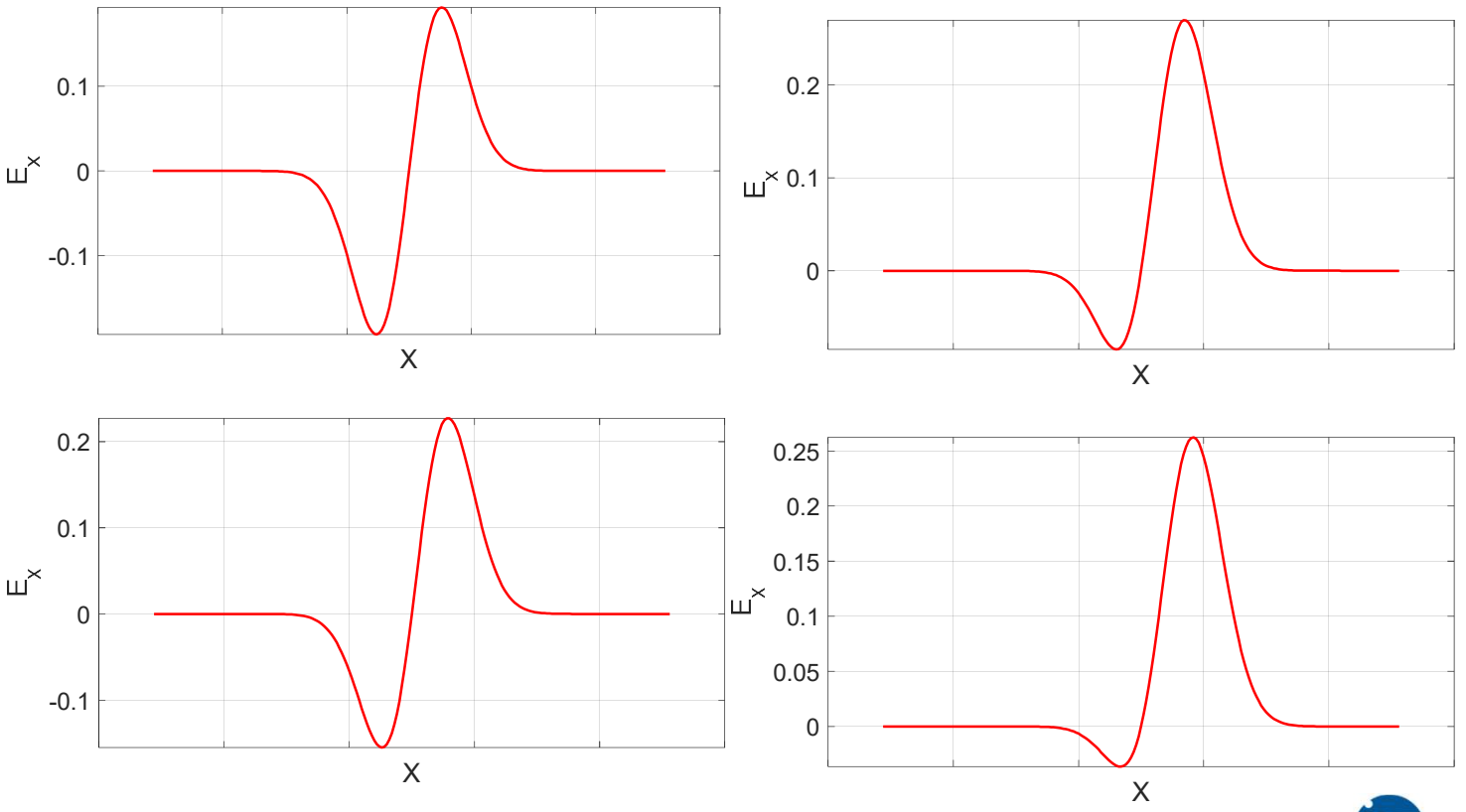
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C_μ In dB

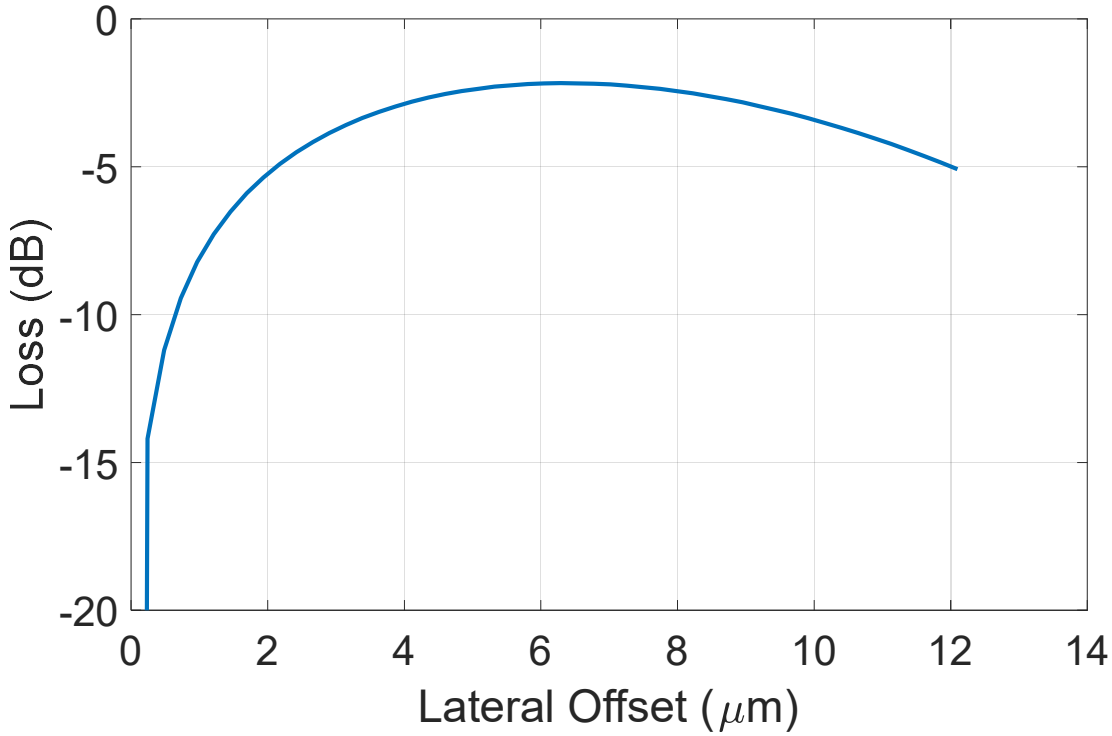
Butt-coupling of SI MMFs: with lateral offset

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Butt-coupling of SI MMFs: with lateral offset

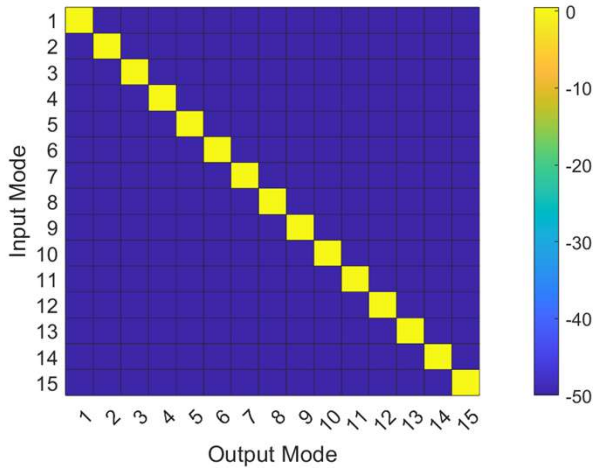
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Germany



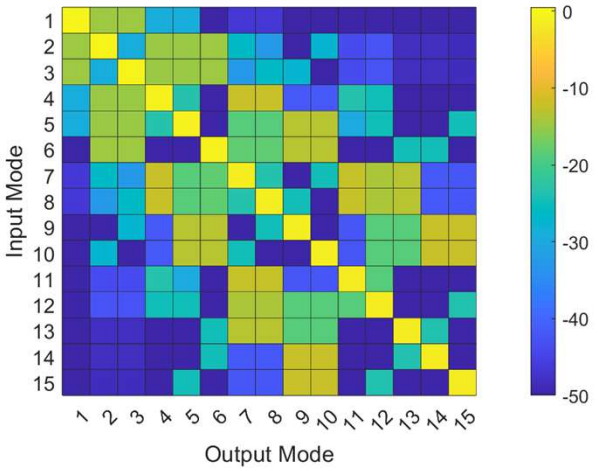
Simulation: Crosstalk vs. offset

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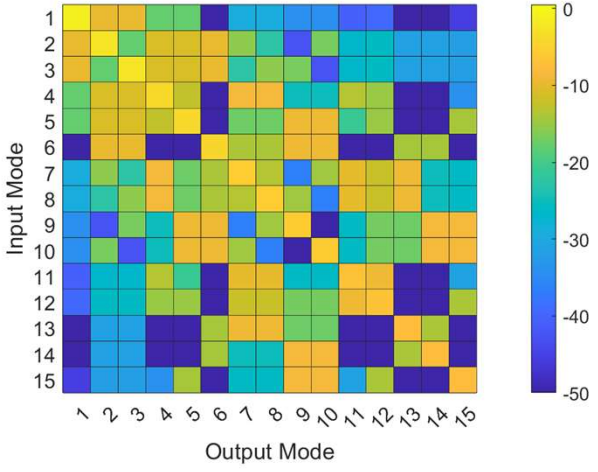
- 0mu m



- 1mu m

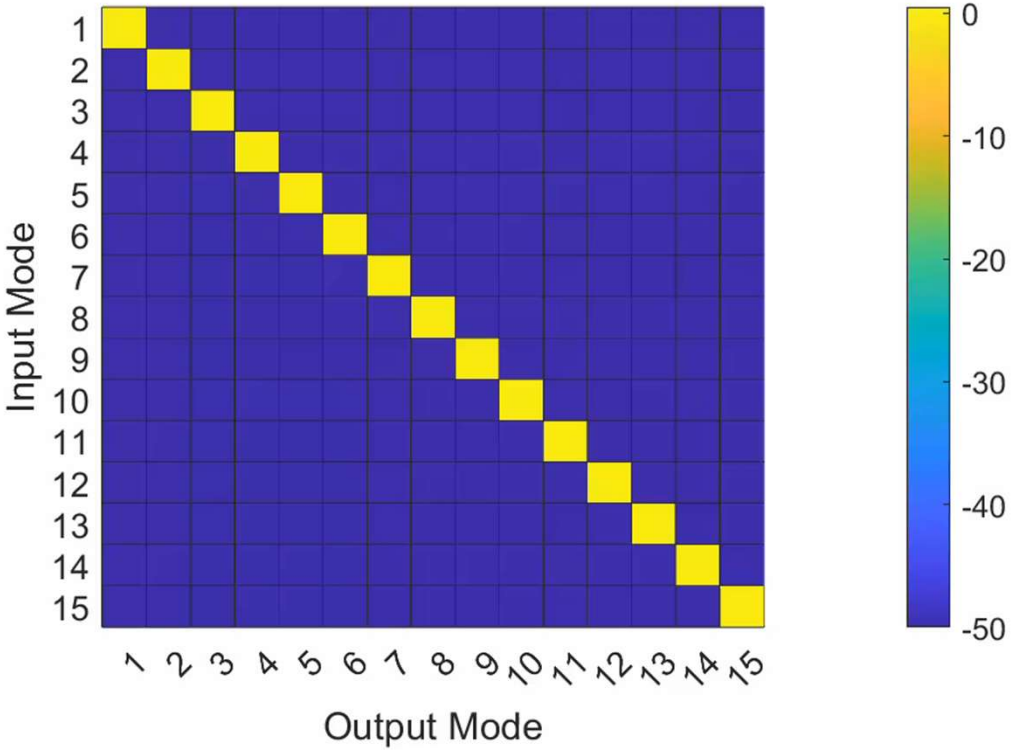


- 3mu m



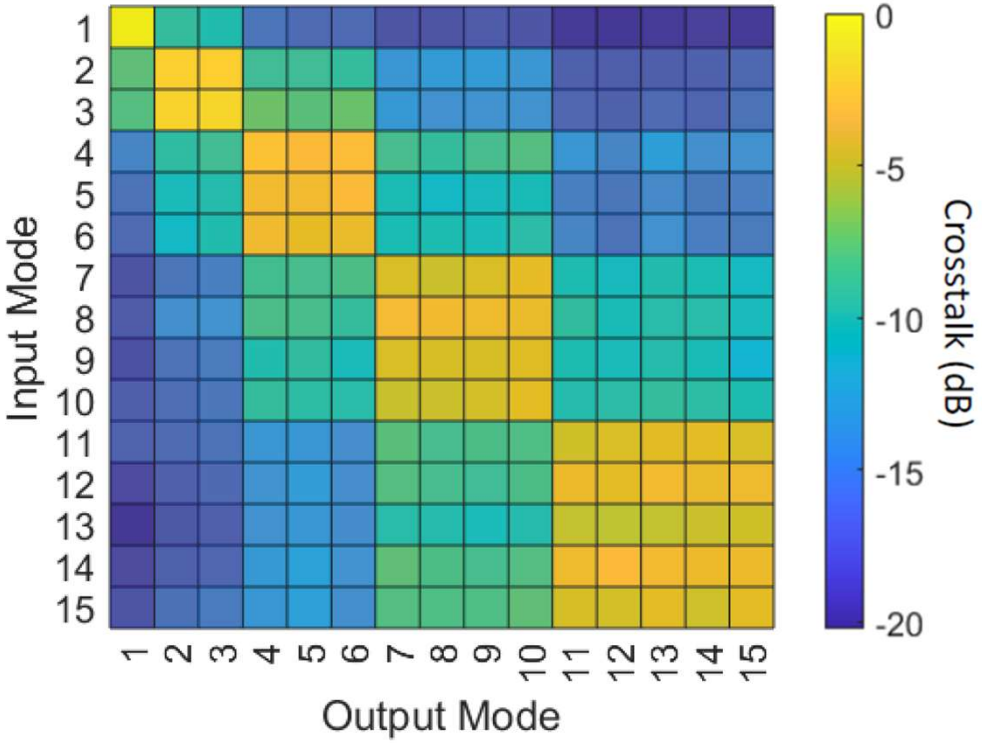
Simulation: Crosstalk vs. offset

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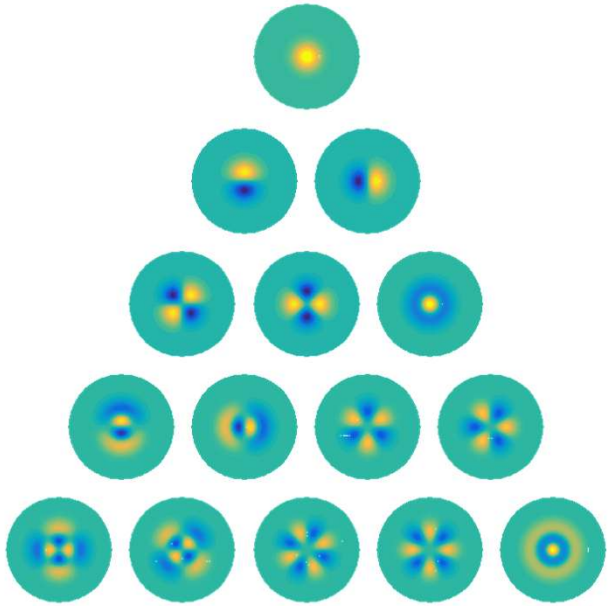




Experiment: Crosstalk



University of Stuttgart
Germany



University of Stuttgart
Institute of electrical and optical communications

Thank you!



Prof. Dr.-Ing. Georg Rademacher

e-mail georg.rademacher@int.uni-stuttgart.de
phone +49 (0) 711 6856 7901
www. int.uni-stuttgart.de/

University of Stuttgart
Institute of electrical and optical communications

